



THE UNIVERSITY OF
TENNESSEE
KNOXVILLE

COSC 594 Special Topics in Computer Science
Biometrics
Fall 2026

Course Section: 003

Meeting Time and Place: 4:05 – 5:20 PM; Min H. Kao 525

Course Credit Hours: 3 credits

Faculty Contact Information

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Course Description/Information:

Graduate seminar on modern biometric recognition and AI. Topics include face, fingerprint, iris, gait, person re-identification, feature representations, similarity learning, verification and identification, evaluation methodologies, robustness, fairness, privacy, and security. Students read foundational and contemporary literature and complete project-based assignments using modern computer vision and machine learning techniques.

Value Proposition:

Beyond the scope of the application of biometrics, this course helps students establish and enhance skills that are necessary for disciplined research—especially relating to Computer Vision, Artificial Intelligence, and Machine Learning.

Student Learning Outcomes/Objectives:

Learning Objectives	Project 1	Project 2	Project 3
1. Formulate biometric recognition problems	✓	✓	✓
2. Develop biometric representations	✓	✓	✓
3. Design & evaluate biometric systems	✓	✓	✓
4. Critique & reproduce biometric research		✓	✓
5. Investigate challenging conditions		✓	✓
6. Conduct & Communicate reproducible research	✓	✓	✓

✓ **Emphasized:** Learning outcome is explicitly developed, practiced, and assessed.

✓ **Developed:** Learning outcome is incorporated

Programmatic Outcomes/Department Goals:

This course aligns departmental goals to teach, train, mentor, and cultivate scholarly excellence in research and education. This course also aligns with the Foundational Artificial Intelligence Cluster, Future Mobility Cluster, and Resilient Agriculture and Forest Systems (RAFS) Cluster.

Learning Environment:

This course will comprise of both in-class and outside-of-class activities. In-class activities will mostly include the introduction of ideas, topics, and theme for biometrics; discussions centered around research papers; and instructor support project work. Outside-of-class activities will include not only course projects, but also participation activities relating to reading assignment discussions using the course [Discord](#) server (*see Course Communications*)

Course Communications:

The primary modes for general announcements, communication and discussion related to course activities will be through the course [Discord](#) server: *COSC 594 - Biometrics (UTK EECS)*.

There are several channels for targeted engagement.

These channels and their specific purpose include:

- **announcements:** official course announcements, reminders, and instructor communications,
- **community:** an informal space for the course community,
- **paper-discussion:** discussion of the assigned weekly papers,
- **project-work:** developing and discussing the major course projects,
- **questions-and-help:** general questions about course concepts, methods, and other topics related to the course.

How to Be Successful in This Course:

Success in this course requires consistent engagement rather than last-minute preparation. Complete the assigned readings before class, participate actively in discussions (online and in-class), and use class time to ask questions, experiment, and collaborate with your peers. Treat the projects as research investigations rather than simply programming assignments. Students should maintain organized code, meaningful Git commits, experiment logs, and clear documentation throughout the process. You are not expected to know everything at the outset—**curiosity, persistence, willingness to experiment, and learning from failed approaches are essential parts of doing research**. When you encounter difficulties, use Discord to ask questions and help your classmates, and seek instructor guidance when needed.

AI Use Policy:

Generative AI tools are permitted in this course as learning and research aids, including for brainstorming, programming assistance, debugging, literature exploration, editing, and other appropriate uses. However, students remain fully responsible for the accuracy, originality, and integrity of all submitted work. AI-generated content, code, or ideas should be *critically evaluated* and *iteratively refined* by the student rather than submitted without review.

When AI tools make a substantive contribution to coursework, students must document and cite their use. Documentation should identify the tool used, the general purpose of its use, and, when relevant, how the student evaluated, modified, or incorporated its output. For project work, this documentation may be included in the project's README, research notebook, or other appropriate project documentation rather than submitted as a separate assignment.

Students may NOT represent AI-generated work as their own without disclosure, fabricate results or citations using AI, or use AI tools in ways that violate course or University academic-integrity policies. Students are ultimately held accountable for everything they submit, including AI-assisted work.

Texts & Reading Materials:

Assigned readings will include journal, conference, and other publicly accessible scholarly literature.

Required Equipment & Resources:

- **Computing Environment:** Students are expected to have access to a computer capable of running a modern Python-based machine learning environment. A personal computer with a reasonably recent CPU, sufficient memory, and adequate storage is sufficient for most coursework.
- **GPU Computing:** A dedicated GPU is recommended but not required. Many of the course projects will involve deep learning and computer vision workloads for which GPU acceleration is highly beneficial. Students without access to a suitable GPU may use the University of Tennessee's ISAAC high-performance computing environment or other instructor-provided/shared computing resources.
- **Software:** Course projects will primarily use Python and common scientific computing and machine learning libraries, including PyTorch and related computer vision tools. Students will also use Git for version control and will be expected to maintain reproducible software environments and documentation.
- **Remote/HPC Access:** Students using ISAAC or other remote computing resources should be comfortable working in a Linux environment and accessing systems through a terminal/SSH. Instruction and starter materials will be provided as needed; prior HPC experience is not required.
- **Other Equipment:** No specialized biometric sensors or hardware are required. Course projects will primarily use publicly available datasets, existing models, and software-based experiments.

Course Resources:

- **Canvas** will be used for sharing grading/assessments, grading policy, and this syllabus.
- **Discord** will be used communication and discussion between (i) the instructor and students and (ii) students with their peers/classmates.
- **Reading Assignments** will utilize UTK's library resources, such as access to databases, journals, and other texts which be found here: <https://lib.utk.edu/>
- **Instructor** will be available for *one-on-one* or *group* meetings outside of class. Meetings can be scheduled via e-mail to determine agreeable times.

Course Requirements, Assessments, and Evaluations:

Grading Scale

A+	≥ 97.0%
A	93.0% to < 97.0%
A-	90.0% to < 93.0%
B+	87.0% to < 90.0%
B	83.0% to < 87.0%
B-	80.0% to < 83.0%
C+	77.0% to < 80.0%
C	73.0% to < 77.0%
C-	70.0% to < 73.0%
D+	67.0% to < 70.0%
D	63.0% to < 67.0%
D-	60.0% to < 63.0%
F	< 60.0%

Course Grade Weighting

project 1	20%
project 2	20%
project 3	35%
participation	10%
practice/reproducibility	15%
	100%

Major Assignments (names and due dates)

- **Project 1**—build a baseline biometrics recognitions system;
Due Sept. 11, 2026 by 11:59pm
- **Project 2**—reproduce a method from a modern biometric journal or conference paper and investigate
Due Oct. 23, 2026 by 11:59pm
- **Project 3**—comprehensive capstone research that
 - identifies a problem,
 - formulates a potential solution,
 - provides reproducible experiments and results*Due Dec 1, 2026 by 11:59pm*
- **Participation**—Students are expected to actively engage in the intellectual life of the course through in-class discussions, research paper discussions, and asynchronous discussion through Discord. Participation includes asking thoughtful questions, contributing ideas, critically engaging with others' perspectives, and helping peers work through technical or conceptual challenges. The goal is *meaningful and consistent engagement*, rather than simply the quantity of contributions.
- **Practice/Reproducibility**—Students will develop professional practices for conducting and communicating computational research. Throughout the course, students are expected to use version control (Git), maintain meaningful commit histories and changelogs, document experiments and computational environments, and organize code and data so that their work can be understood and reproduced by others. These practices will be evaluated primarily through project repositories, research logs, documentation, and other project deliverables. The emphasis is on producing clear, organized, and reproducible research rather than on the number of commits or amount of documentation.

Course Feedback:

Course feedback and suggestions may be provided through anonymously through course evaluations, through the community discord channel.

Key Campus Resources for Students:

- [Center for Career Development and Academic Exploration](#) (Career counseling and resources; Handshake job search system)
- [Course Catalogs](#) (Listing of academic programs, courses, and policies)
- [Hilltopics](#) (Campus and academic policies, procedures and standards of conduct)
- [OIT HelpDesk](#) (865) 974-9900
- [Schedule of Classes/Timetable](#)
- [Student Health Center](#) (visit the site for a list of services)
- [Academic Success Center](#) (Academic support resources)
- [Undergraduate Academic Advising](#) (Advising resources, course requirements, and major guides)
- [University Libraries](#) (Access to library resources, databases, course reserves, and services)

Course Schedule:

Week	Day	Date	Topic
Part I — Foundations of Biometric Recognition			
1	Tuesday	8/18/26	What Does It Mean to Recognize a Person?
	Thursday	8/20/26	
2	Tuesday	8/25/26	What Should a Biometric Representation Look Like?
	Thursday	8/27/26	
3	Tuesday	9/1/26*	How Should We Learn Identity?
	Thursday	9/3/26*	
Part II — Measuring Recognition			
4	Tuesday	9/8/26	How Do We Know Whether a Biometric System Is "Good?"
	Thursday	9/10/26	
5	Tuesday	9/15/26	How Should Biometric Experiments Be Designed?
	Thursday	9/17/26	
6	Tuesday	9/22/26	Can We Trust Published Biometric Results?
	Thursday	9/24/26	
Part III — Recognition Under Distribution Shift			
7	Tuesday	9/29/26	Why Do Biometric Systems Fail Outside the Laboratory?
	Thursday	10/1/26	
8	Tuesday	10/6/26†	Can We Make Representations Generalize?
	Thursday	10/8/26	
9	Tuesday	10/13/26	What Should a Model Do When It Encounters Something It Has Never Seen?
	Thursday	10/15/26	
Part IV — Modern AI for Biometrics			
10	Tuesday	10/20/26	Do Foundation Models Change Biometric Recognition?
	Thursday	10/22/26	
11	Tuesday	10/27/26	Can Different Biometric Modalities Help Each Other?
	Thursday	10/29/26	
12	Tuesday	11/3/26	How Do We Make Biometrics Work in Difficult Conditions?
	Thursday	11/5/26	
Part V — Security, Privacy, and Responsible Biometrics			
13	Tuesday	11/10/26	Can Biometric Systems Be Fooled?
	Thursday	11/12/26	
14	Tuesday	11/17/26	Can We Build Biometrics Responsibly?
	Thursday	11/19/26	
Part VI — Research			
15	Tuesday	11/24/26	What important research questions remain?
	Thursday	11/26/26‡	
16	Tuesday	12/1/26	Biometrics Research Symposium

* Asynchronous (no in-person class)

† Fall break (no in-person class)

‡ Thanksgiving break (no in-person class)

The instructor reserves the right to revise, alter or amend this syllabus as necessary. Students will be notified in advance by email and via Discord of any such changes.

The Campus Syllabus

Dear Student,

The purpose of this **Campus Syllabus** is to provide you with important information that applies to all UTK courses. Please observe the following policies and familiarize yourself with the university resources listed below. At UT, we are committed to providing you with a high-quality learning experience. I want to wish you the best for a successful and productive semester.

–Dr. John Zomchick, Provost and Senior Vice Chancellor

ACADEMIC INTEGRITY

Each student is responsible for their personal integrity in academic life and for adhering to UT's Honor Statement. The Honor Statement reads: "An essential feature of the University of Tennessee, Knoxville is a commitment to maintaining an atmosphere of intellectual integrity and academic honesty. As a student of the university, I pledge that I will neither knowingly give nor receive any inappropriate assistance in academic work, thus affirming my own personal commitment to honor and integrity."

YOUR ROLE IN IMPROVING THE COURSE THROUGH ASSESSMENT

At UT, it is our collective responsibility to improve the state of teaching and learning. During the semester you may be requested to assess aspects of this course, either during class or at the completion of the class, and through the TNVoice course evaluation. Please take the few moments needed to respond to these requests as they are used by instructors, department heads, deans and others to improve the quality of your UT learning experience.

STUDENTS WITH DISABILITIES – <http://sds.utk.edu>

The University of Tennessee, Knoxville, is committed to providing an inclusive learning environment for all students. If you anticipate or experience a barrier in this course due to a chronic health condition, a learning, hearing, neurological, mental health, vision, physical, or other kind of disability, or a temporary injury, you are encouraged to contact [Student Disability Services](#) (SDS) at 865-974-6087 or sds@utk.edu. An SDS Coordinator will meet with you to develop a plan to ensure you have equitable access to this course. If you are already registered with SDS, please contact your instructor to discuss implementing accommodations included in your course access letter. You may also contact your SDS Coordinator if you have any questions, concerns, or experience any barriers to using your accommodations in this course.

ACCESSIBILITY POLICY AND TRAINING – <http://accessibility.utk.edu>

The University of Tennessee, Knoxville, provides reasonable accommodations for individual students with disabilities through its office of [Student Disability Services](#). The university is also committed to making information and materials accessible, when possible. Resources and assistance to support these efforts can be found at <http://accessibility.utk.edu/>.

WELLNESS – <http://wellness.utk.edu/> and <http://counselingcenter.utk.edu/>

The [Center for Health Education and Wellness](#) empowers all Volunteers to thrive by cultivating personal and community well-being. The Center can answer questions about general wellness, substance use, sexual health, healthy relationships, and sexual assault prevention. The [Student Counseling Center](#) is the university's primary facility for personal counseling, psychotherapy, and psychological outreach and consultation services.

[The Student Health Center](#) has a wide range of clinics and services. Their goal is to provide quality healthcare and assist you in maintaining a healthy lifestyle to support your success as a student. To schedule an

appointment, call 865-974-5080. To report a health-related absence visit: <https://studentlife.utk.edu/caring-for-all-vols/absence/>.

Any student who has difficulty affording hygiene products, groceries, or accessing sufficient food to eat every day is urged to contact [Basic Needs](#) for support. The Big Orange Pantry, located on the 3rd floor of the Student Union, is a free resource for all students, faculty, and staff, no matter how great or small their need is. Students who need emergency financial assistance can also request funding from the [Student Emergency Fund](#).

Students who are experiencing non-academic difficulty or distress and need assistance should call 974-HELP or [submit an online referral](#). The 974-HELP team specializes in aligning resources and support to students experiencing mental health distress.

EMERGENCY ALERT SYSTEM – <http://safety.utk.edu/>

The University of Tennessee, Knoxville, is committed to providing a safe environment for learning and working. When you are alerted to an emergency, please take appropriate action. Learn more about what to do in an emergency and sign up for UT Alerts. Check the emergency posters near exits and elevators for building specific information. In the event of an emergency, course schedules and assignments may be subject to change. If changes to graded activities are required, reasonable adjustments will be made, and you will be responsible for meeting revised deadlines.